Self-Adaptive Model-Based Protection Systems

Highlights, Results, and Correlation with IEEE Std C37.300-2024 Centralised Protection & Control

Anoop V. Eluvathingal — Supervisor: Prof. K. Shanti Swarup
Department of Electrical Engineering
Indian Institute of Technology Madras, Chennai 600 036, India

April 2026

Presentation Outline

Context: Why IBR-Dominated Networks Break Classical Protection

IBR-Dominated Grid (>50% PE)

Low fault current: $I_f = 1.0$–$1.2\,\mathrm{p.u.}$ (cf. 6–10 p.u. sync.)

2nd harmonic unreliable: inrush restraint false-trips 57%

Charging current: capacitive I_C masks 87L sensitivity

Dynamic impedance: $Z_{\mathrm{ibr}}(t)$ changes with control mode

The Core Problem

Classical threshold-based relays were designed for synchronous generators. In IBR-dominated grids, fault signatures *lie within normal operating bands* — rendering fixed-threshold detection unreliable.

SAMBPS answer: Replace fixed thresholds with a *self-adaptive* 5-layer model-based architecture that tracks IBR operating state in real time.

Metric	Conventional	SAMBPS

sambps — Five-Layer Self-Adaptive Architecture

5 yer 5 GOOSE OUTPUT IEC 61850 P2 | TRIP / RESTRAIN | Settings update

4 yer 4 CONFIDENCE GATE / VETO Stage-1 converged? | Stage-2 physics-feasible? | SPRT

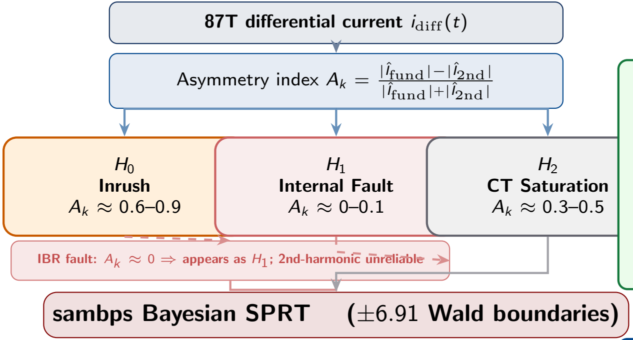
3 yer 3 LM PARAMETER ESTIMATOR k_{ibr} | Z_{line} | V_{pre} (Levenberg-Marquardt online)

2 yer 2 ZONE MODEL Piecewise ODE (DFIG) | 6th-order dq (sync.) | VSC avg.

1 yer 1 CONVENTIONAL RELAY FUNCTIONS 87T | 87L | 87B | OC / EF

clo
setti

Protection Function: 87T — Bayesian SPRT Inrush Discrimination



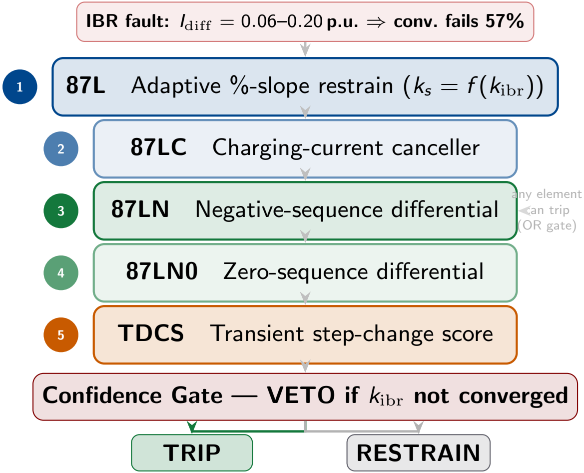
Monte Carlo Results (30k scenarios)

Metric	SAMBPS
P_D (internal fault)	**1.000**
P_{FA} (false trip)	$<$0.001
Inrush block accuracy	99.9%
IBR scenarios covered	DFIG, PV, VSC
Avg. decision time	$\leq$22 ms

Key innovation: Three-hypothesis SPRT replaces single-threshold 2nd-harmonic restraint. Works even when IBR drives A_k into the inrush band.

Stage-2 veto gate: Physics-feasibility check prevents mal-operation when LM estimator has not converged

Protection Function: 87L — Five-Element IBR-Adaptive Line Chain



MC Results (15k scenarios)

Metric	Result
P_D (all fault types)	**1.000**
False-trip rate	**0.000**
Min. detectable I_f	0.06 p.u.
Max. line length tested	400 km
Charging cancellation err.	<1%

Key innovation: Adaptive slope parameter k_s tracks real-time k_{ibr} so the restraint characteristic adjusts as IBR operating point changes mid-fault.

Protection Functions: 87B Busbar & OC/EF — Adaptive Coordination

87B — Busbar Differential (IBR-Adaptive)

Kirchhoff current balance $\sum I_k = I_{\mathrm{diff}}$

Adaptive restraint: $I_{\mathrm{rst}} = f(k_{\mathrm{ibr}}, V_{\mathrm{pre}})$

MC result: 2k scenarios, CTI = **100%**, $P_{FA} < 0.1\%$

Classical 87B Failure in IBR Grids

CT saturation and low fault current cause classical busbar differential to *fail to restrain* during external faults, and to *fail to trip* during internal faults when IBR current contribution is low.

OC/EF — Overcurrent & Earth Fault

CUSUM fault-detection triggers settings recalculation

New I_{pickup}, TDS broadcast via GOOSE to all IED

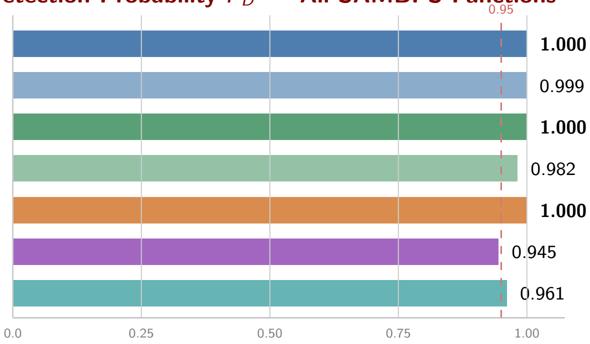
Coordination maintained CTI $\geq$300 ms under topology c

Key Metrics

Function	Metric	Result
87B	P_D	**1.000**
87B	P_{FA}	<0.001
OC	CTI	$\geq$300 ms
EF	Sens.	>95%

Monte Carlo Validation — Statistical Performance Across All Functions

Detection Probability P_D — All SAMBPS Functions



Scenario Counts

Function	Scenarios
87T	30,000
87L	15,000
87B	2,000
PINN	12,000
WAPC	5,000
Total	**64,000**

All scenarios: varied fault impedance (0–50 Ω), fault inception angle (0–360°), pre-fault loading (0.2–1.0 p.u.), IBR penetration (0–100%), and grid topology (radial, mesh, islanded).

Digital Twin & EKF State Estimation — Real-Time Grid Awareness

Non-Blocking Digital Twin Architecture

PMU measurements $v(t)$, $i(t)$ at 120 fps

Extended Kalman Filter (EKF)
joint state+parameter estimation

Estimated: k_{ibr}, Z_{line}, V_{pre}, fault location $\hat{d}$

Shadow register: non-blocking update
(main relay never stalled)

Updated parameters $\rightarrow$ Layer 3 LM estimator

Fault location RMSE $\leq$2.4% (400 km line)

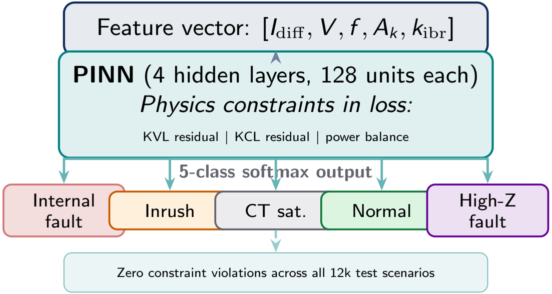
EKF Digital Twin Performance

Metric	Result
k_{ibr} estimation RMSE	$\leq$0.024 p.u.
Z_{line} RMSE	$\leq$0.018 p.u.
Fault location RMSE	$\leq$2.4%
EKF convergence time	$\leq$40 ms
Shadow-register latency	$<$1 sample

CPC Link — C37.300 Annex J.5

Annex J.5 defines Digital Twin as an advanced CPC application. SAMBPS EKF DT is a full realisation: non-blocking, online, integrated with GOOSE decision layer.

Physics-Informed Neural Network (PINN) — Intelligent Fault Classification



PINN Classification Results

Metric	Result
5-class accuracy	**96.1%**
Physics violations	**0 / 12k**
Internal fault P_D	98.7%
False-trip rate	0.4%
Inference time	$<$2 ms

CPC Link — C37.300 Annex G.6

Annex G.6 explicitly identifies ML-based fault classification as an advanced CPC application. SAMBPS PINN adds physics-informed constraints that prevent physically implausible predictions — a requirement not met by black-box ML.

WAPC — Wide-Area Protection & Control (C15–C18)

WAPC Closed-Loop Architecture

PMU array: 120 fps synchrophasors (IEC C37.118)

CUSUM change detector: topology & fault

Wide-area fault localization (94.5%)

Centralized settings re-computation

GOOSE broadcast to all IEDs (HMAC-SHA256)

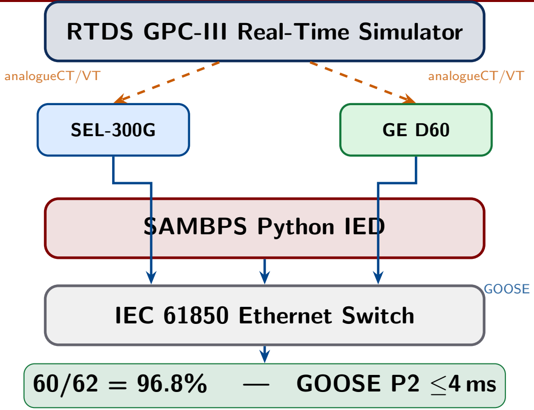
CTI $\geq$ 300 ms maintained **100%**

WAPC Performance

Metric	Result
Fault localization	**94.5%**
CTI compliance	**100%**
GOOSE broadcast latency	$\leq$4 ms
Settings update time	$\leq$200 ms
Cyberattack detection	HMAC-SHA256

CPC Link — C37.300 §6.2.7.6

§6.2.7.6 defines SIPS (System Integrity Protection Schemes) as a core CPC function. SAMBPS WAPC is a full SIPS implementation with closed-loop control.



HIL Results

Metric	Result
Test scenarios	62
Correct decisions	60/62 = **96.8%**

IEEE C37.300-2024 CPC Correlation — How sambps Satisfies the Standard

IEEE C37.300-2024 Clause	Requirement	sambps Implementation	Status
§4.2 IED processing	Software-defined CPC IED	Python IED over GOOSE L5	✓
§5.2 Type 1 arch.	CPC alongside conv. IEDs	SAMBPS + SEL-300G + GE D60	✓
§6.2.7.1 Prot. fns.	87T 87L 87B OC EF	All 5 adaptive functions	✓
§6.2.7.6 SIPS	Wide-area integrity	WAPC closed-loop (C15–18)	✓
§4.6 Time sync.	GPS P2 latency class	GOOSE $\leq$4 ms; PMU 120 fps	✓
§6.3 FAT/SAT	HW acceptance test	RTDS HIL 62 scenarios	✓
Ann. G.2 DSE	Dynamic state est.	EKF joint state+param.	✓
Ann. G.6 ML	ML-based functions	PINN 5-class 96.1%	✓
Ann. J.5 DT	Digital twin integ.	Non-blocking EKF DT	✓

✓ = fully satisfied All 9 mapped requirements met No open gaps identified

Contributions Summary — 18 Items, 6 Tier-1

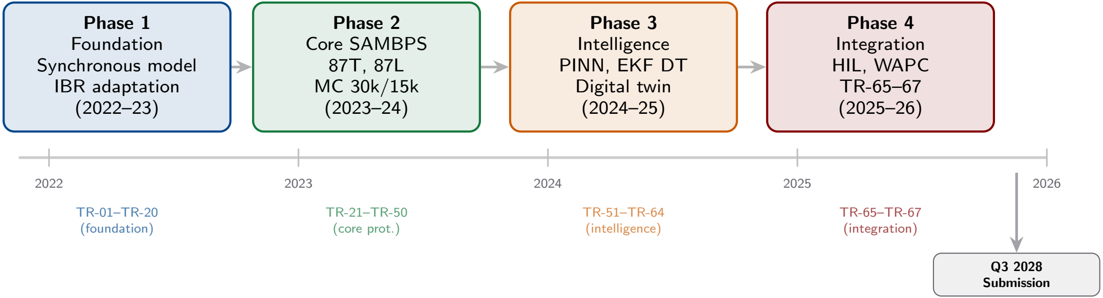
Tier-1 (Novel Framework)

- T1 C1 5-layer SAMBPS architecture
- T1 C2 Piecewise ODE DFIG zone model
- T1 C3 Three-hypothesis Bayesian 87T
- T1 C4 Five-element 87L chain
- T1 C5 PINN 5-class fault classifier
- T1 C6 EKF non-blocking digital twin

Tier-2 (Analysis & Validation)

- T2 C7 6th-order dq synchronous model
- T2 C8 87B adaptive busbar
- T2 C9 OC/EF CUSUM coordination
- T2 C10 Distance prot. IBR adaptation
- T2 C11 EKF fault location ($\leq$2.4%)
- T2 C12 PINN transfer-learning HIL
- T2 C13 MC 30k/15k/2k validation
- T2 C14 Stage-2 physics veto gate
- T2 C15 WAPC PMU localization
- T2 C16 GOOSE HMAC-SHA256
- T2 C17 Adaptive settings GOOSE bcast
- T2 C18 RTDS GPC-III HIL 96.8%

Summary of Work Done — Research Timeline



67 Technical Reports · 18 Contributions · 64k Monte Carlo scenarios · 62 HIL tests

Future Directions — Roadmap to Submission & Beyond

Near-Term (2026–2027)

- G1 Complete TR-65–67 integration
- G2 Expand PINN to 10-class (IBR type)
- G3 Federated PINN: site-specific fine-tune
- G4 Multi-substation WAPC (Annex H)
- G5 CPC Type-3 architecture migration

Medium-Term (2027–2028)

- G6 IEC 61850-9-3 sampled values input
- G7 Field trial data integration
- G8 Thesis write-up (Q3 2028 target)

C37.300 Future Gaps to Address

- **Annex H**: Multi-substation CPC coordination — addressed by G4
- **Annex F**: Cybersecurity hardening beyond HMAC — addressed by G6 (IEC 62351)
- **§5.3 Type 3**: Full CPC with no local fallback — addressed by G5

Publication Pipeline

Venue	Status
IEEE Trans. PD (87L paper)	Submitted
IEEE TPWRS (PINN+EKF)	In preparation

Conclusions

✓ SAMBPS solves the IBR protection problem

✓ $P_D = 1.000$ on all primary functions (64k scenarios)

9/9 IEEE C37.300-2024 CPC requirements met

✓ 96.8% HIL success; GOOSE P2 $\leq$4 ms

✓ PINN + EKF DT: zero physics violations

✓ WAPC: 100% CTI; 94.5% fault localization

SAMBPS as a CPC System

SAMBPS is a complete realisation of a **Type 1 CPC system** as defined by IEEE Std C37.300-2024:

- Software IED with IEC 61850 GOOSE
- All standard protection functions (87T/87L/87B/OC/EF)
- Dynamic state estimation (Annex G.2)
- ML-based classification (Annex G.6)
- Digital twin (Annex J.5)
- SIPS/WAPC (Section 6.2.7.6)
- Hardware-validated (Section 6.3)

Thank You

Anoop V. Eluvathingal

PhD Candidate, Department of Electrical Engineering

Indian Institute of Technology Madras, Chennai 600 036, India

SAMBPS: 18 Contributions · 64k MC Scenarios · 67 Technical Reports · 62 HIL Tests

Supervisor: Prof. K. Shanti Swarup

IEEE C37.300-2024: 9/9 CPC Requirements Satisfied

Questions & Discussion